

ACADEMIC PROJECT PROPOSAL

Uflash

A Secure Super-App for University Students and Staff

Managing Food, Travel, and Safety Using a Unique Uni-Code

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1. Abstract

Uflash is a proposed mobile Super-App designed to serve as a comprehensive ecosystem for university students and staff. The platform integrates food ordering with a Smart-Queue system, ride-hailing via Pick Me API, AI-powered allergy safety screening, and crowdsourced campus navigation. By optimizing daily workflows, Uflash aims to save each student an average of 20 minutes per day. All features are secured under a Uni-Code identity verification system.

2. Introduction

University campuses are dynamic environments where members navigate daily challenges such as long canteen queues, unreliable transport, and food allergy risks. Existing solutions are fragmented. Uflash proposes a unified, campus-exclusive platform that consolidates these services into one secure interface. By leveraging the Uni-Code system, only verified members are granted access, ensuring a safe and trusted environment.

3. Problem Statement

Key pain points identified include:

- **Time Loss:** Long physical queues at canteens during peak hours resulting in wasted time and missed classes.
- **Navigation & Cost:** Commercial GPS tools lack the granularity to identify specific campus gates or lecture hall entrances. Hiring survey companies for mapping typically costs approximately LKR 200,000 and becomes outdated quickly.
- **Health Safety:** Risks of allergic reactions due to inadequate ingredient transparency at food outlets.
- **Transport Safety:** Unreliable or unsafe transport options for commuting to and from campus.
- **Campus Community:** Lack of a unified identity platform that fosters community and healthy competition.

4. Proposed Solution — Uflash

Uflash addresses these challenges through a single, verified platform with five integrated modules.

4.1 Food Categories & Management (Flash-Order & Flash-Queue)

- **University Canteens (Internal):** Uses a Smart-Queue (Digital Token) system allowing users to place orders in advance and skip physical lines.
- **Flash-Queue (Monetized):** Users can pay LKR 20 (or use a Pro membership) to use a priority counter, effectively monetizing impatience to fund operations.

- **Local Canteens (External):** Includes local shops and home-cooks listed on the platform to provide greater variety beyond the standard campus menu.

4.2 Health Safety: AI Allergy Guard

This intelligent tool scans ingredients and cross-checks them against the user's personal health profile to prevent accidental allergen exposure. While included for Pro members, it is available for an LKR 50 daily pay-per-use fee.

4.3 Travel & Crowdsourced Smart Maps

- **Flash-Ride ("Let's Go"):** Integrated ride-hailing via the Pick Me API, offering safe, perimeter-controlled travel for campus commuters.
- **Smart Maps (The Wikipedia Approach):** A specialized GPS layer showing precise university gates and lecture hall positions.

How it Works: Students "add or fix" locations themselves by dropping a pin and adding a photo. The location goes live once confirmed by three other students.

Cost Saving: By utilizing student contributions instead of a survey company, Uflash saves LKR 200,000 and ensures the map is completed in one week.

Incentives: Students participate for free to earn "Campus Map King" badges or free Flash-Queue skips.

4.4 Social & Gaming: Uni-Arena

A gamified social layer where students play mini-games to earn points and "pride scores" for their university. This fosters engagement and rewards users with free food coupons, serving as a zero-cost viral marketing tool.

5. Membership & Revenue Model

Revenue Stream	Fee / Rate	Details & Impact
Joining Fee	LKR 100	One-time activation (~LKR 2 lakh per 2,000 users)
Student Pro	LKR 300 / sem	Unlocks Premium Outside Canteens, AI Guard, and Arena
Staff Pro	LKR 500 / sem	Full feature access for university staff
Uni Cafes	Fixed Fee	Periodic fee for canteens joining the Smart-Queue system
Local Canteens	Monthly Fee	Listing fee to access the campus market
Ride Commission	Reg + 10%	Registration fee plus commission per ride booked

6. Technology Stack & Operational Efficiency

- **Backend:** Python is used for the AI Allergy Guard engine and API bridges.
- **Security:** Every account is verified via the Uni-Code to keep the platform exclusive.
- **Cost Saving:** Servers utilize an auto-shutdown mechanism after peak hours, reducing hosting bills by 80%.
- **Zero Ad Budget:** Viral growth is achieved through Uni-Arena competition and peer-to-peer mapping rewards.

7. Conclusion

Uflash is a holistic, campus-first digital solution that addresses real daily challenges. By combining crowdsourced data, smart server scaling, and diverse revenue streams, the platform is designed to be profitable from Semester 1. Uflash is a campus ecosystem designed to help students and staff save time, stay safe, and stay connected.

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